

Assignment 2 Part 1 Report

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Algorithm Description

I used Kruskal's Algorithm to determine the Minimum Spanning Tree (MST). This calculates the lowest cost to connect all stations. As required, I implemented a custom Disjoint Set structure using Partition, Cluster, and Node classes as per previous assignment. Instead of a standard array-based parent pointer, each Cluster maintains a Linked List of Node objects. To ensure efficiency and avoid timing errors, I implemented "Weighted Union" (Union by Size). When merging two clusters, the smaller linked list is appended to the larger linked list, and the pointers of the smaller list are updated. This keeps the Union operation efficient. The edges are sorted using Java's Collections.sort (MergeSort/TimSort variant), which operates in $O(E \log E)$.

Implementation Details

Input is handled using BufferedReader and StringTokenizer instead of Scanner to handle large inputs quickly. A HashMap<String, Integer> is used to map the station names to integer IDs for array indexing. The problem asks if Peter can reach all stations. In MST terms, if the algorithm successfully adds (N-1) edges (where (N) is the number of stations), the graph is connected. If fewer edges are added, the graph is disconnected, and the output is "Impossible".

Testing & Status

The program was tested locally against three sample inputs, the first input was provided in the assignment and it produced the exact sample output. The other two custom inputs are as follows:

Input1.txt:

```
3 2
A
B
C
A B 10
B C 20
A
3 1
A
B
C
A B 10
A
```

```

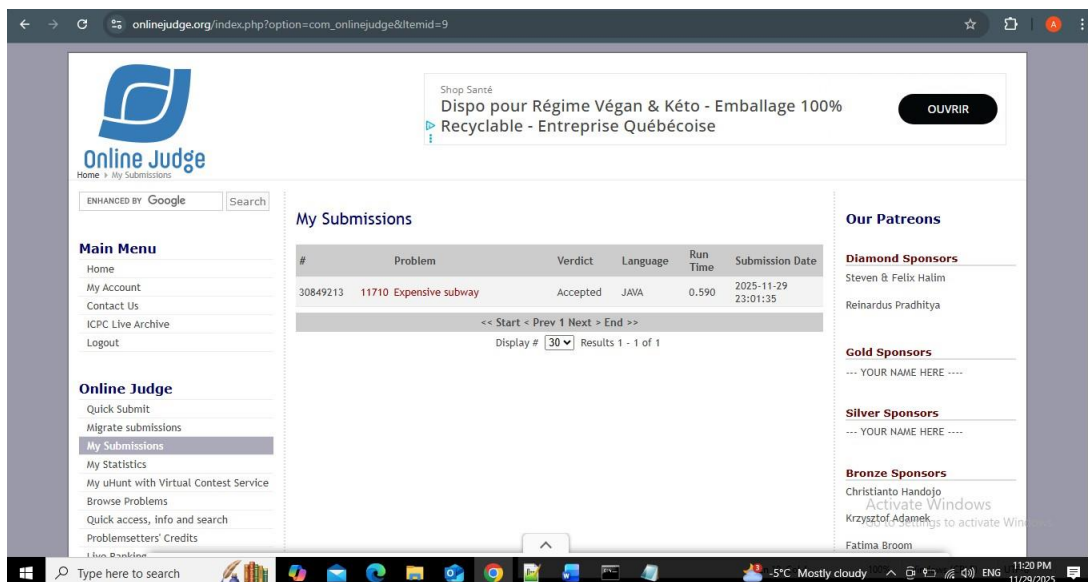
1 0
LonelyStation
LonelyStation
0 0

Input2.txt:
4 3
Central
North
South
Isolated
Central North 10
North South 15
Central South 20
Central
    0 0

```

I submitted the code to UVa Online Judge. The verdict showed as accepted and the runtime is 0.590s

As per the last tests, there are no known bugs. The solution handles the corner case of a single station correctly by outputting 0.



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5608]
(c) Microsoft Corporation. All rights reserved.

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>javac Main.java

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < input.txt
12
Impossible

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < input1.txt
30
Impossible

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < input2.txt
Impossible

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>
```

After unzipping the assignment files, I found six extra tests for part one. I run them one by one and here are the results:

```
C:\Windows\System32\cmd.exe
53( Belleville - Place des Fêtes ) - $121
54( Châtelet - Odéon ) - $122
55( Bastille - Hôtel de Ville ) - $127
56( Gare d'Austerlitz - Place d'Italie ) - $169
57( Michel Ange Auteuil - Trocadéro ) - $204

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < test1.txt
12
Impossible

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < test2.txt
45

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < test3.txt
Impossible

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < test4.txt
Impossible

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < test5.txt
0

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>java Main < test6.txt
45

D:\uOttawa\3rd term\CSI\CSI Assignment\Programming Assignment 2>
```